



Python for Data Science - 2305CS303

Lab - 8

Roll No. : 122

Name : Yatharth Sanghavi

01) Write a function to calculate BMI given mass and height.
($BMI = mass/h^{**2}$)

```
In [13]: def bmi(mass,height):  
          return mass/(height**2)  
  
          mass=100  
          height=2  
          print(bmi(mass,height))
```

25.0

02) Write a function that add first n numbers.

```
In [5]: def addn(n):  
         return sum(range(1,n+1))  
         addn(10)
```

Out[5]: 55

03) Write a function that returns 1 if the given number is Prime or 0 otherwise.

```
In [11]: def prime(n):  
          for i in range(2,n):  
              if n%i==0:  
                  return 0  
          else:  
              return 1  
          r=prime(11)  
          if r==1:  
              print("number is prime")
```

```
else:  
    print("number is not prime")
```

number is prime

04) Write a function that returns the list of Prime numbers between given two numbers.

```
In [15]: def prime(n,n2):  
          result=[]  
          for i in range(n,n2+1):  
              for j in range(2,i):  
                  if i%j==0:  
                      break  
              else:  
                  result.append(i)  
          return result  
n=1  
n2=10  
result=prime(n,n2)  
result
```

Out[15]: [1, 2, 3, 5, 7]

05) Write a function that returns True if the given string is Palindrome or False otherwise.

```
In [20]: def palindrome(s):  
          return s == s[::-1]  
r=palindrome("nayan")  
r
```

Out[20]: True

06) Write a function that returns the sum of all the elements of the list.

```
In [21]: def suml(l):  
          return sum(l)  
l=[1,2,3,4,5]  
r=suml(l)  
r
```

Out[21]: 15

07) Write a function to calculate the sum of the first element of each tuples inside the list.

```
In [33]: l=[(1,2,3),(1,2,3),(1,2,3)]  
          r=0  
          for i in range(len(l)):  
              r+=l[i][0]  
          r
```

Out[33]: 3

08) Write a recursive function to find nth term of Fibonacci Series.

```
In [99]: def fibonacci(n):
          l=[0,1]
          for i in range(n):
              p=l[-1]
              pn=l[-2]
              r=p+pn
              l.append(r)
          return l[n]
          n=5
          print(fibonacci(n))
```

5

09) Write a function to get the name of the student based on the given rollno.

Example: Given dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'} find name of student whose rollno = 103

```
In [48]: dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'}
          rollno = 103
          print(dict1.get(rollno))
```

Jay

10) Write a function to get the sum of the scores ending with zero.

Example : scores = [200, 456, 300, 100, 234, 678]

Ans = 200 + 300 + 100 = 600

```
In [60]: def sum_endzero(scores):
          r=0
          for i in scores:
              if i%10==0:
                  r+=i
          return r
          scores = [200, 456, 300, 100, 234, 678]
          print(sum_endzero(scores))
```

600

11) Write a function to invert a given Dictionary.

hint: keys to values & values to keys

Before : {'a': 10, 'b':20, 'c':30, 'd':40}

After : {10:'a', 20:'b', 30:'c', 40:'d'}

```
In [62]: d= {'a': 10, 'b':20, 'c':30, 'd':40}
r={v:k for k,v in d.items()}
r
```

```
Out[62]: {10: 'a', 20: 'b', 30: 'c', 40: 'd'}
```

12) Write a function that returns the number of uppercase and lowercase letters in the given string.

example : Input : s1 = AbcDEfgh ,Ouptput : no_upper = 3, no_lower = 5

```
In [66]: s1 = "AbcDEfgh"
no_upper=0
no_lower=0
for i in s1:
    if i.isupper():
        no_upper+=1
    elif i.islower():
        no_lower+=1
print("no_upper=",no_upper)
print("no_lower=",no_lower)
```

```
no_upper= 3
```

```
no_lower= 5
```

13) Write a lambda function to get smallest number from the given two numbers.

```
In [72]: n=10
n2=5
r=(lambda n,n2:n if n<n2 else n2)(n,n2)
print(r)
```

```
5
```

14) For the given list of names of students, extract the names having more that 7 characters. Use filter().

```
In [87]: student=['rahul','ram','ravan','krish','sachin','ramanandan']
r= list(filter(lambda s:len(s)>=7,student))
r
```

```
Out[87]: ['ramanandan']
```

15) For the given list of names of students, convert the first letter of all the names into uppercase. use map().

```
In [90]: student=['rahul','ram','ravan','krish','sachin','ramanandan']
r=list(map(lambda s:s.title(),student))
r
```

```
Out[90]: ['Rahul', 'Ram', 'Ravan', 'Krish', 'Sachin', 'Ramanandan']
```

16) Write udfs to call the functions with following types of arguments:

1. Positional Arguments
2. Keyword Arguments
3. Default Arguments
4. Variable Length Positional(*args) & variable length Keyword Arguments (**kwargs)
5. Keyword-Only & Positional Only Arguments

In [117...

```
def Positional(a,b,c):
    print(a,b,c)
def Keyword(a,b,c):
    print(a,b,c)
def Default(c):
    print(c)
def VariableLP(*c):
    print(c)
def VariableLK(**k):
    print(k)
def keywordo(*,a,b,c):
    print(a,b,c)
def positionalo(a,b,c,/):
    print(a,b,c)

Positional(1,2,3)
Keyword(2,b=3,c=1)
Default(4)
VariableLP(1,2,3,4)
VariableLK(a=10,b=20,c=20)
keywordo(a=0,b=0,c=0)
positionalo(1,2,3)
```

```
1 2 3
2 3 1
4
(1, 2, 3, 4)
{'a': 10, 'b': 20, 'c': 20}
0 0 0
1 2 3
```